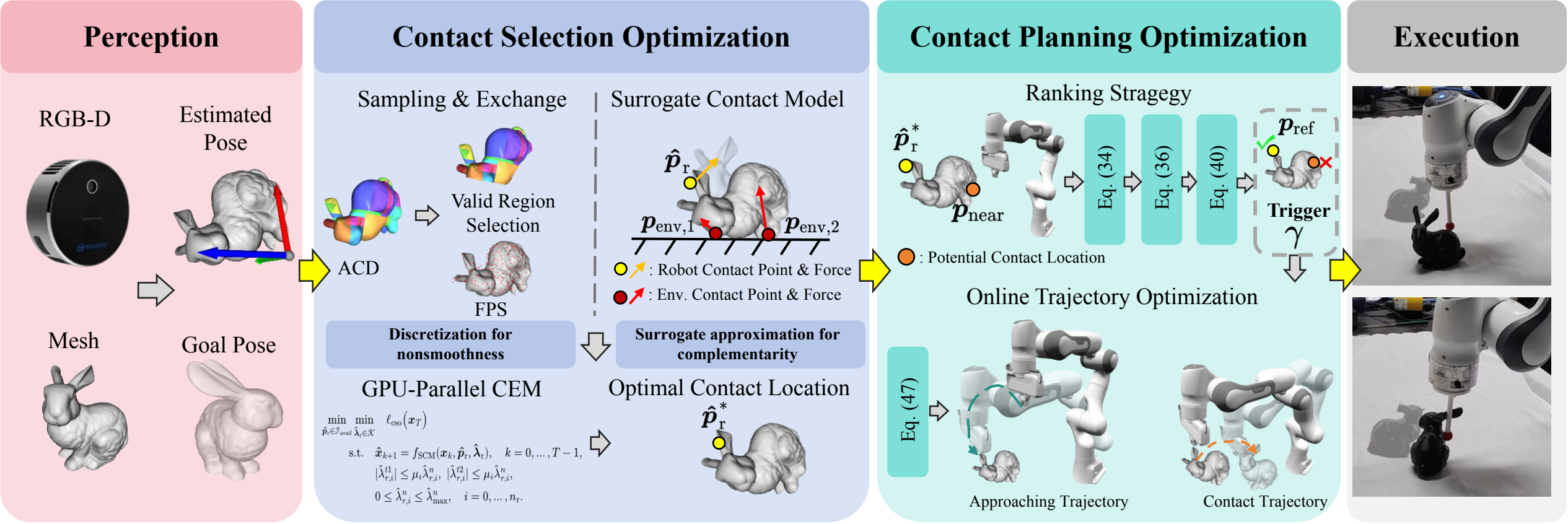

Perception
RGB-D
Estimated Pose
Mesh
Goal Pose
Contact Selection Optimization
Sampling & Exchange
Valid Region Selection
ACD
FPS
Surrogate Contact Model
$\hat{p}_r$
$p_{env,1}$
$p_{env,2}$
: Robot Contact Point & Force
: Env. Contact Point & Force
Discretization for nonsmoothness
Surrogate approximation for complementarity
GPU-Parallel CEM
$\min_{\hat{p}_r \in \mathcal{J}_{avail}} \min_{\hat{\lambda}_r \in \mathcal{K}} \ell_{cso}(x_T)$
s.t. $\hat{x}_{k+1} = f_{SCM}(x_k, \hat{p}_r, \hat{\lambda}_r), \quad k = 0, \dots, T-1,$
$|\hat{\lambda}^{t1}_{r,i}| \le \mu_i \hat{\lambda}^n_{r,i}, \ |\hat{\lambda}^{t2}_{r,i}| \le \mu_i \hat{\lambda}^n_{r,i},$
$0 \le \hat{\lambda}^n_{r,i} \le \hat{\lambda}^n_{max}, \quad i = 0, \dots, n_r.$
Optimal Contact Location
$\hat{p}^*_r$
Contact Planning Optimization
Ranking Stragegy
$\hat{p}^*_r$
p_{near}
Eq. (34)
Eq. (36)
Eq. (40)
p_{ref}
Trigger
γ
: Potential Contact Location
Online Trajectory Optimization
Eq. (47)
Approaching Trajectory
Contact Trajectory
Execution